

B.3 Best-of-n Sampling

We train the reward model using a pairwise loss:

$$\mathcal{L}_{RM} = -E_{(x, y_w, y_l) \sim D} [\log \sigma(r(x_i, y_w) - r(x_i, y_l))] \quad (4)$$

where (x, y_w, y_l) is a question-chosen-rejected triplet from the preference dataset. During training of the reward model, we set batch size to 128, learning rate to 9e-6, and train for one epoch. We partition 10% of the preference data to construct a validation set to select the best checkpoint. We use 4 A100 80G GPUs for reward model training.

B.4 Proximal Policy Optimization

We employ the SFT model and reward model obtained from B.3 for PPO training. We use DeepSpeed-Chat⁶ for PPO training. The SFT model and reward model used in PPO training are obtained from the BoN’s supervised fine-tuning and reward modeling. For PPO (Schulman et al., 2017), the loss function of the actor model is:

$$\mathcal{L}_{PPO-Actor} = -\hat{E}_t[\max(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t), r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)} \quad (5)$$

And the loss function of the critic model is:

$$\mathcal{L}_{PPO-Critic} = 0.5 * \hat{E}_t[\max((V_\phi(s_t) - \hat{R}_t)^2, \text{clip}(V_\phi(s_t), V_{old}(s_t) + \epsilon, V_{old}(s_t) - \epsilon))] \quad (6)$$

We set the learning rate for both the actor model and the critic model to 1e-6. The generation batch size is 64 and the training batch size is 32. Each training step, we train a single inner epoch. We utilize DeepSpeed ZeRO-3 to train one epoch on 32 A100 80G GPUs.

B.5 Hindsight Instruction Relabeling

We combine 10 Idk datasets using the HIR method, constructed from 10 distinct Ik thresholds ranging from 0.1 to 1.0. These Ik thresholds correspond to knowledge confidence level from 1.0 to 0.1, respectively. The lower the knowledge confidence level, the less confident the model is in its own knowledge, resulting in a more conservative response strategy. Besides, we also add a dataset consisting entirely of refusals to respond, corresponding to situations where the knowledge confidence level is 0 and its Ik threshold can be seen as 1.1. We utilize the following formula to convert from the Ik threshold to the knowledge confidence level:

$$\text{Knowledge_confidence_level} = 1.1 - \text{Ik_threshold} \quad (7)$$

For example, we prepend the following instruction to questions in the Idk dataset corresponding to an Ik threshold of 1.0:

Your current knowledge expression confidence level is 0.1, please answer the user’s question: <Question>

We set the batch size to 256, the learning rate to 2e-5 and we train for 3 epochs using 8 A100 80G GPUs. The advantage of this method is that it allows users to control the model’s response strategy without the need to retrain the model. For instance, in scenarios where there is a low tolerance for factual errors, we can set the knowledge confidence level to 0.1. This setting prompts the model to answer only those questions it is particularly certain about, thereby ensuring truthfulness. Conversely, in situations where there is a higher tolerance for factual errors, we can adjust the knowledge confidence level to 1.0. This adjustment encourages the model to respond to a wider range of questions, enhancing its helpfulness. We show the comparison between Idk-HIR and Idk-SFT in Appendix C.3.

⁶<https://github.com/microsoft/DeepSpeedExamples/tree/master/applications/DeepSpeed-Chat>

C Additional Experimental Results

C.1 Knowledge Quadrants Under Different Ik Thresholds

In Figure 7, we present the distribution of the model’s knowledge quadrants after Idk-SFT when the Ik threshold ranges from 0.1 to 0.9.

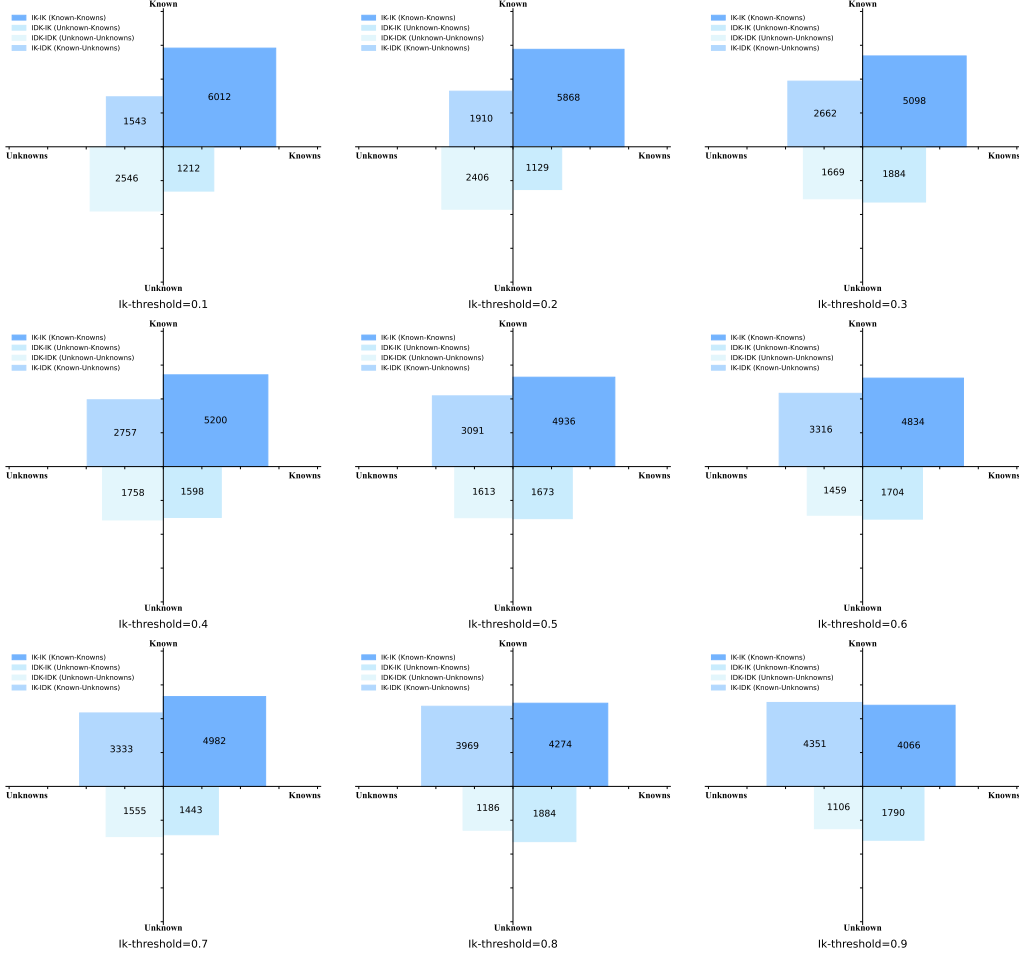


Figure 7: Knowledge quadrants under different Ik thresholds.

C.2 Effect of IK threshold

Answer F1 and Refusal F1. We report Answer F1 score and Refusal F1 score to reflect changes in the model’s behavior influenced by the Ik threshold. Regarding Answer F1, we only consider whether the model answer the question, without taking into account the accuracy of the answer.

As shown in Figure 8, when the Ik threshold raises, the model tends to refuse to answer questions, resulting in an increase in Refusal F1. Conversely, when the Ik threshold is low, the model is more inclined to answer questions, leading to an increase in Answer F1.

C.3 Idk-HIR vs Idk-SFT

In this section, we compare the effects of Idk-HIR and Idk-SFT.

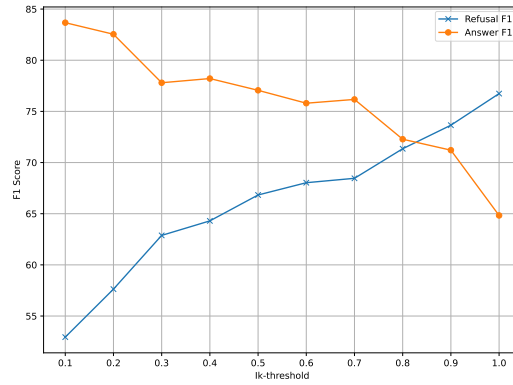


Figure 8: Refusal F1 and Answer F1 scores at different Ik thresholds.

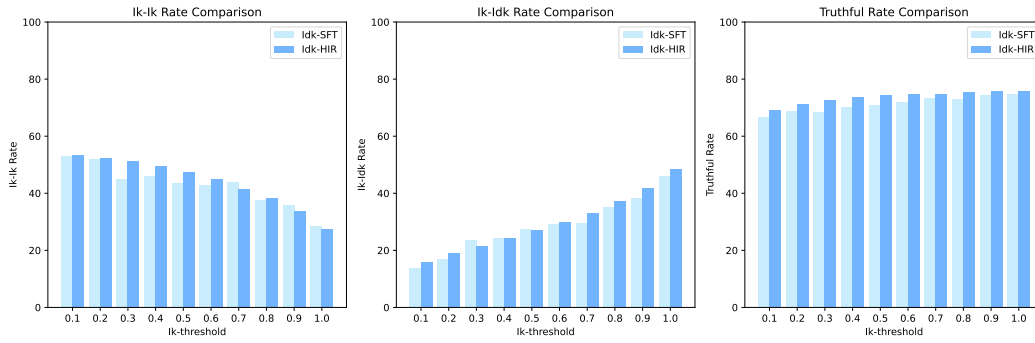


Figure 9: Comparison between Idk-SFT and Idk-HIR.

As shown in Figure 9, the IK-IK rate and IK-IDK rate of the Idk-HIR model are comparable to those of the Idk-SFT model across various Ik thresholds, and the TRUTHFUL rate is consistently higher than that of the Idk-SFT. Therefore, in certain scenarios, the flexible and controllable Idk-HIR model serves as an excellent alternative to the Idk-SFT model.